

Assignment#2 : Measures of Central Tendency, Dispersion, and Box Plot

1. A factory measured the tensile strength (in MPa) of 10 steel rods:
430, 445, 455, 435, 440, 450, 455, 460, 435, 445
 - Compute:
 - a) Mean strength
 - b) Median strength
 - c) Mode strength
 - Briefly interpret which measure is more appropriate in this context and why.
2. For the same tensile strength data above:
 - Calculate:
 - a) Range
 - b) Variance
 - c) Standard Deviation
 - Comment on the consistency of the material based on the results.
3. A) Draw a **box plot** for the tensile strength data.
 - Clearly label all five-number summary components.
 - Comment if there's any outlier and what that might mean in an engineering production setting.
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B) Draw the box plot For the following Dataset 1 and dataset 2, and then compare.

Dataset 1

Values: 50, 10, 1, 5, 7, 8, 9, 10, 12, 15, 100

Dataset 2

Values: -50, -10, 1, 5, 7, 8, 9, 10, 12, 15, 100

4. The following data represents the **resistance (in ohms, Ω)** of 50 resistors measured in a lab.
 - a) **What is the mean** (Arithmetic mean, Geometric mean, and Harmonic mean) resistance of the 50 resistors measured in ohms (Ω)?
 - b) **What is the median** resistance of the 50 resistors measured in ohms (Ω)?
 - c) **What is the modal** resistance of the 50 resistors measured in ohms (Ω)?
 - d) Compute the coefficient of variation.

Table : Resistance (Ω) of 50 Resistors Measured in a Laboratory

Class Interval (Ω)	Frequency (f)
95 – 99	5
100 – 104	9
105 – 109	11
110 – 114	13
115 – 119	7
120 – 124	3
125 – 129	2